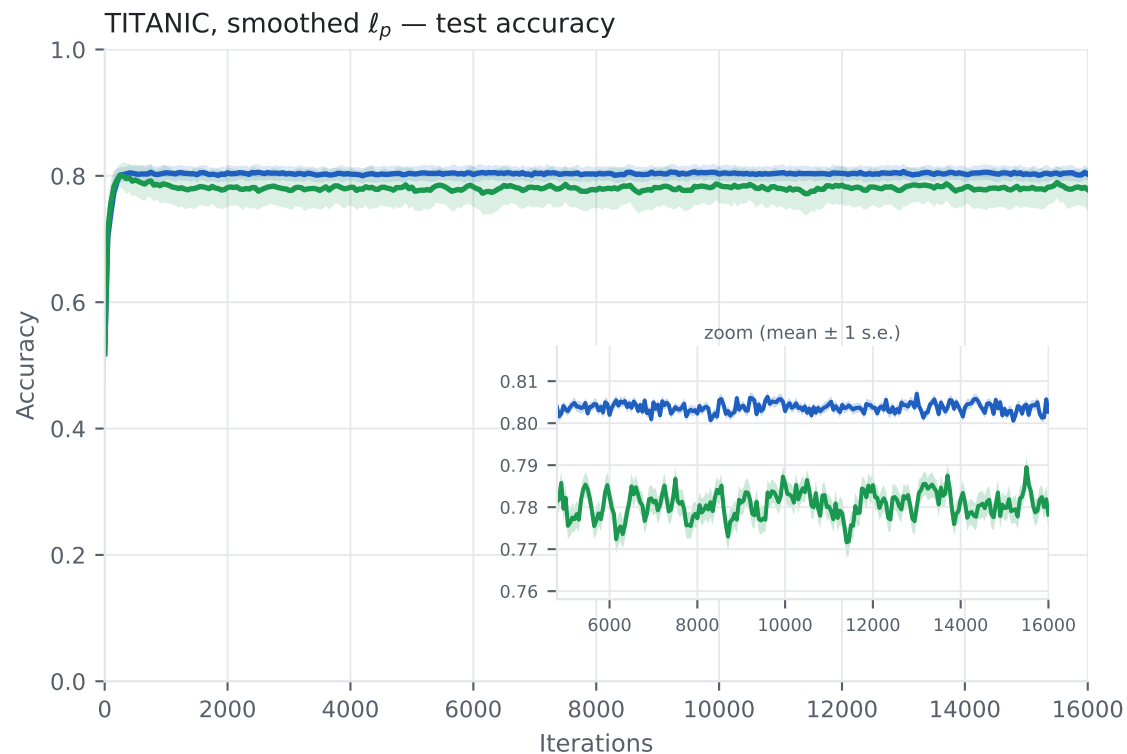
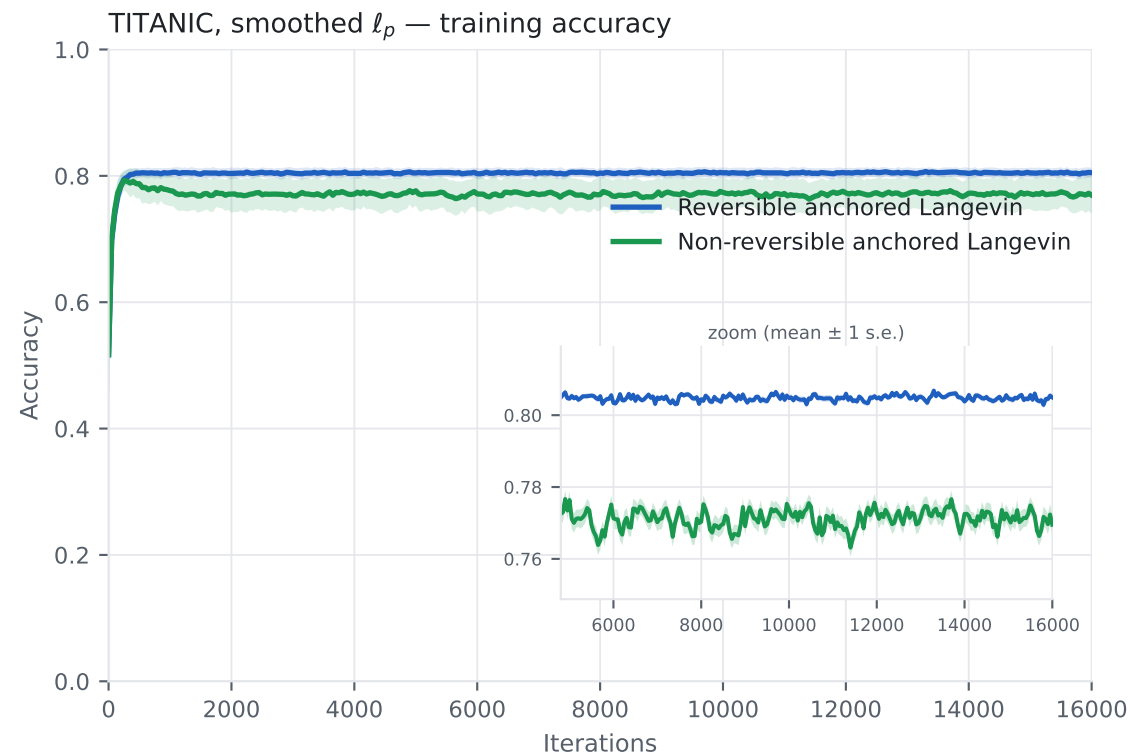


On this geometry the same s that buys mixing costs accuracy



Titanic, smoothed ℓ_p , $p = 2.4$, $\text{eps} = 0.18$, $\text{Lambda} = 4$, $d = 10$, exact gradient, LASSO anchor ($\text{sigma} = 10$, $\text{lambda} = 7.12$, $\text{delta} = 0.0108$, a in $[0.5, 1]$). EXACTLY the configuration of the mixing figures: $\text{eta} = 0.0001$, $s = (5, 5, 5)$, $R = 96$ coupled chains, 16000 iterations, seed 5300. Paired final-iterate difference (non-reversible minus reversible): training -0.0355 ($t = -12.35$), test -0.0245 ($t = -6.96$); averaged over all post-burn-in checkpoints instead of the final one, -0.0336 (training) and -0.0228 (test). The non-reversible arm is clearly WORSE here -- 2.3 percentage points of test accuracy -- and that is the price of the same $s = 5$ that buys the mixing gain: on this geometry J 's axis is $\text{grad } g$ rather than w , so the same strength produces a roughly 3x stronger rotation and a correspondingly larger discretisation bias. The mixing win on this set is real but it is not free. Bands are mean $\pm$ 1 SD across replicates; the inset shows mean $\pm$ 1 s.e. so the curves can be separated at all. The two arms share an invariant law, so their accuracy distributions are identical at stationarity -- this figure is what that looks like, and no tuning changes it.